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Atharva Uday Kalase

Computer Science Engineer

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Objective

Data scientist and MSc Data Science & AI candidate at Queen's University Belfast, with over two years' professional experience spanning satellite-imagery research at ISRO's Space Applications Centre and enterprise software delivery at Avalara. Published in peer-reviewed remote sensing venues and named on an issued Indian patent, with a track record of taking computer-vision systems from research question to validated prototype — including a rooftop solar-potential model at 93% accuracy across three Indian cities and a real-time ambulance detection system deployed on edge hardware at 98% accuracy. Seeking a data science or AI engineering role where Multimodal and geo-spatial machine learning delivers measurable operational impact.

Education

M.S. in Data Science & Artificial Intelligence, *Queens University, Belfast, UK* Jan 2026 - Jan 2027
Dissertation: Visual Question Answering for Remote Sensing Imagery — QLoRA fine-tuning of Qwen2.5-VL with Grounding DINO for object grounding, benchmarked on VRSBench.
Modules: Machine Learning, Artificial Intelligence, Data Mining, Data Analytics, Data Visualization, AI Ethics.

B.Tech in Electronics & Communication Engineering, *Vishwakarma Institute of Technology, Pune* Aug 2019 - May 2023
Academic CGPA: 8.65/10 || Percentage: 80%
Modules: Data Structures, Machine Learning, Computer Vision, Geographic Information Systems, Signal Processing

Professional Experience

Implementation Consultant **Mar 2024 - May 2025**
Avalara, Pune **Full Time**

- Owned end-to-end implementation of tax automation software for a portfolio of 150+ enterprise customers, from ERP and Salesforce integration through configuration, go-live, and post-launch support.
- Diagnosed and resolved integration, calculation, and returns-filing defects across the AvaTax suite, holding configurations to audit-ready standards; placed 3rd across India and the US in the Q3 2024 random quality audit.
- Recognition: Bull's Eye — Best Throughout (Jul & Dec 2024, Jan & Feb 2025); Most Valuable Performer (Jan & Feb 2025); Orange Ambassador of Avalara (2025–26).
- **Tech Stack:** AvaTax, Salesforce, Jira, Gainsight, SQL, US sales tax & returns filing.

Computer Vision and GIS Researcher **Jul 2022 - Jul 2023**
Indian Space Research Organization, Space Application Center, Ahmedabad **Full-time**

- Served in the Advanced Visualization and Development Department under the guidance of Mr. Jai Singla (Sci / Engineering, Space Application Center, ISRO).
- **Projects included:** SolView solar potential analysis, Urban Visibility Analysis, and Fort Visibility Analysis.
- **Tech stack:** Python, Machine Learning, Computer Vision, Software Development, Research & Development, & GIS.

Skills

Tech Stack (Familiar):	C and C++ Programming, SQL Server, Python Programming, Software Development, Machine Learning, Artificial Intelligence, IOT (Internet of Things), Cloud Technologies, and Linux.
Domains:	Operating Systems, Computer Vision, Statistical Analysis, Deep Learning, Data Visualization, NLP, Object Detection, Remote Sensing, Geographic Information Systems, Troubleshooting, Performance Analysis, Quality Testing, Product Development, and Automation.
Soft Skills:	Research and Development, Project Leadership, Teamwork, Written and Verbal Communication Skills, Competitive Thinking, Problem-Solving, Adaptability, Creativity, and Time Management.

Projects

EmergiSense: Multi-modal Ambulance Detection System

- Developed a multimodal deep learning system that detects approaching ambulances at traffic signals from audio and video, cutting emergency response time by **30+ seconds at 98% accuracy.**

- Deployed inference on **NVIDIA Jetson Nano** for real-time roadside operation in Python, PyTorch, and OpenCV; **one national patent issued.**
- Led the team to **Rank 1 nationwide** at the Smart India Hackathon; pitched to Maharashtra Police, AICTE and the BHAU Institute Cell, COEP Pune.

SolView: Solar Potential Analysis and Satellite Imagery and DEM
- Developed software for rooftop solar potential analysis using high-resolution DEM and satellite imagery (achieved 93% accuracy over major Indian cities).
- **Tech Stack -**Python, OpenCV, ML, ArcGIS Pro, Remote Sensing, Computer Vision, and Software Development.
- Achieved an **accuracy of 93%** over Ahmedabad, Gandhinagar and Delhi. This can impact solar application efficiency.

RSVQA — Visual Question Answering for Remote Sensing Imagery
- Building a vision-language system that answers natural-language questions about satellite imagery — object counting, presence, relative position, and land-use reasoning — so non-specialists can interrogate Earth observation data without GIS training.
- Fine-tuning Qwen2.5-VL with QLoRA for parameter-efficient adaptation on a single GPU, and pairing it with Grounding DINO so answers are anchored to located objects rather than generated unsupported.
- Stack: Python, PyTorch, Hugging Face Transformers, vision-language models, PEFT/QLoRA, open-vocabulary object detection, remote sensing.

Certifications

- Artificial Intelligence in Power Electronics- Electric Car — Jun 2021
- Range Over Rocketry Webinar - ISRO VSSUT Space Innovation Centre — Jun 2021
- Python for Everybody - University of Michigan — Aug 2020
- The Data Scientist's Toolbox - Johns Hopkins University — Sep 2020
- C++ : Zero to Hero in 2020 - Udemy - Ankit Thakran — Aug 2020

Achievements

Professional Achievements
- **Entrepreneurship Excellence Award 2025** by Entrepreneurship Startup and Innovation Forum under the Student Ambassador Category.
- **Smart India Hackathon Winner (Team Lead):** Led the team to secure **Rank 1 nationwide** in the prestigious **Smart India Hackathon 2022 and received INR 1 Lakhs.**
- **Best Presentation Award:** Received at the International Conference on Innovation in Computer Science, Electrical and Electronics Engineering **(ICICEEE – 22)** hosted by Anna Saheb Dange College of Engineering, Ashta.

Academic Achievements
1. Published over 3 research papers in Scopus-indexed conferences and journals; one national patent has been issued for the EmergiSense project.
2. Publications include:
   - **"Multimodal Detection of Ambulance Using Jetson Nano" || International Conference on Innovations in Computer Science, Electronics and Electrical Engineering, June 2023**
   - **"Solar Potential Analysis Over Indian Cities Using High-Resolution Satellite Imagery and DEM" || Journal of the Indian Society of Remote Sensing, June 2025**
3. Completed a graduate course **"Introduction to Satellite and Engineering"** by the Scoltech university, Russia & European Space Agency, Europe.

Competition Wins
- Secured **2nd Runner-Up** position in PRATYAGNA 2022, INGENIOUS 22, and PICT INC IMPETUS 2022 and 2023.
- **Received an INR 1,00,000 scholarship** to rank in the **top 15 among 1000 police personnel wards of Fiat India Automobile Pvt. Ltd.** under the CSR scheme.
- Presented "EmergiSense: Multimodal ambulance detection system" to **Maharashtra Pune Police Department.**